Guardrails Mechanisms

1. Rule-Based Guardrails

Predefined Rules & Patterns

(Keywords, Regex, or pattern matching)

Fast, Interpretable, Brittle

NeMo Guardrails

Llama Firewall

Wildflare GuardRail

2. Model-Based Guardrails

Specialized, Small Classifiers

(Non-LLM ML models)

Efficient, Specific Detection

Perspective API

ToxiChat

HateXplain

ToxiGen

3. LLM-Based Guardrails

Leverages LLMs for Evaluation

Nuanced, Flexible, High Latency

Discriminative
(Classifier)

Examples:
LlamaGuard,
ShieldGemma

Generative
(Reasoning Judge)

Examples:
GuardReasoner
R²-Guard

4. Hybrid Guardrails

Layered Strategy

Defense-in-Depth

Rule-Based Filter → Model-Based Classifier → LLM Judge

Palisade

Rebuff

Robust, Comprehensive Coverage